Poisson regression

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Warmup

Work on the warmup activity (handout)

Warmup

$e^{0.0265} \approx 1.027$

	Estimate	Std. Error	z value	Pr(>|z|)	
(Intercept)	0.276493	0.087294	3.167	0.00154	**
phd	0.008071	0.026358	0.306	0.75944	
ment	0.026537	0.002001	13.264	< 2e-16	***
kid5	-0.099696	0.035248	-2.828	0.00468	**

How would we interpret the coefficient on the number of articles published by the student's mentor?

For each additional paper published by the mentor (in last 3 years), there is an estimated increase of about 2.7% in the average # of articles published by the student (in last 3 years), holding PhD prestige & children constant.

Warmup

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What is the predicted number of articles for a student at a PhD program with a prestige score of 4, whose mentor published 30 articles in the last 3 years, and who has no children?

$$\exp\{0.2765 + 0.008071(4) + 0.0265(30)\}$$

$$\approx 3 \text{ articles}$$

EDMs and GLMs

We say that a distribution belongs to the **exponential dispersion model** family if the distribution's probability function can be written in the form

$$f(y;\theta,\phi) = a(y,\phi)\exp\left\{\frac{y\theta-\kappa(\theta)}{\phi}\right\}$$

$$f(y;\mu,\phi) = b(y,\phi)\exp\left\{-\frac{d(y,\mu)}{2\phi}\right\}$$

For iid observations $(\mathbf{x}_1, Y_1), ..., (\mathbf{x}_n, Y_n)$, a **generalized linear model** (GLM) consists of two components:

$$Y_i|\mathbf{x}_i \sim \text{EDM}(\mu_i,\phi)$$

$$g(\mu_i) = \mathbf{x}_i^T\beta$$

What we know so far

GLMs:

- Estimating $\boldsymbol{\beta}$ (and ϕ if needed)
- Key properties of EDMs

Linear and logistic regression:

- Model assessment
- Predictions and prediction error
- Inference
- Model selection

Today: Model assumptions and diagnostics for *Poisson* models

Poisson regression

$$Y_i|\mathbf{x}_i \sim Poisson(\mu_i)$$
$$\log(\mu_i) = \mathbf{x}_i^T \beta$$

Question: What assumptions is this model making? How would we assess those assumptions?

- independence
- shape: log mean is linear X
  - empirical log mean plot
  - randomized quantile residuals
- Distribution: $Y_i|x_i$ is count variable and specifically Poisson (so assuming $Var(Y_i|X_i) \approx \mathbb{E}[Y_i|X_i]$)
  - if model is a good fit, quantile residuals $\approx N(0,1)$
    - ↳ QQ plot of quantile residuals to assess overall model fit
    - ↳ randomized quantile residuals vs. X non-constant variance suggests wrong distribution
- Cook's distance
- VIFs, pairs plots & correlations

Activity

Use the provided packet to answer the following questions:

- Does the Poisson regression model appear to be a good fit?
- If not, which piece of the model do you think is mis-specified?